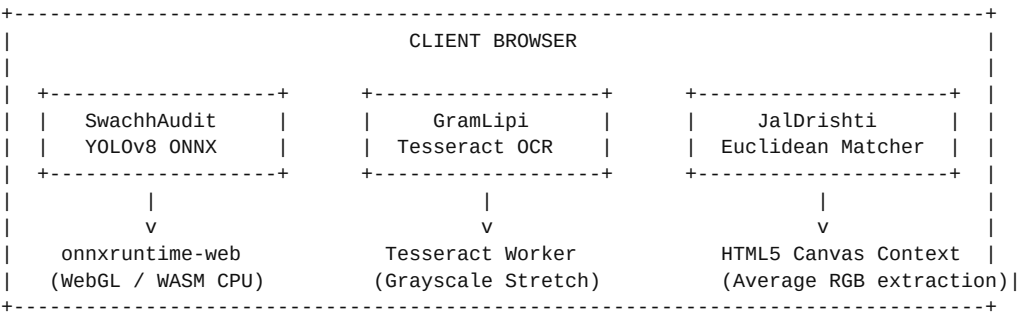


GramSetu AI - Local AI Verification Manual

This document provides verification details for the local, on-device AI implementations in GramSetu AI. It specifies which models run in-browser, when network connectivity is used, and how data privacy is maintained.

1. AI Execution Architecture

GramSetu AI executes machine learning models entirely inside the client's web browser using standard web technologies.



2. On-Device AI Verification Matrix

This matrix details the runtime, execution environment, and network requirements for each feature:

Feature	Runs Locally?	Internet Required?	Model / Algorithm	Runtime Engine	User Data Leaves Device?
Litter Detection (SwachhAudit)	Yes	No	YOLOv8-nano (yolov8n.onnx)	onnxruntime-web (WebGL / WASM)	No
Cleanliness Scoring	Yes	No	Custom bounding box area ratio algorithm	Native JavaScript	No
Notice Text Extraction (GramLipi)	Yes	No	Tesseract OCR (eng .traineddata)	Tesseract.js Web Worker	No
Image Preprocessing	Yes	No	Contrast stretching and grayscale conversion	HTML5 Canvas Context API	No
Text Translation (GramLipi)	Yes	No	Llama-3.2-3B	Local Ollama Server API (localhost)	No (Stays on local host network)
Pad Extraction (JalDrishti)	Yes	No	Coordinate-based guide region crop	HTML5 Canvas Context API	No
Strip Color Matching	Yes	No	Euclidean color distance matching	Native JavaScript	No
Cleanliness Charts (SwachhSankalp)	Yes	No	Custom SVG graphs	React / Virtual DOM	No

3. Data Processing Boundaries

Data That Remains 100% In-Browser

- **Video Frames & Camera Snapshots:** Captures from the device camera are processed in memory and drawn to HTML5 canvas components. No raw images are sent over the network.
- **Extracted OCR Text:** Extracted text from scanned circulars is processed locally in memory.
- **Cleanliness Reports:** The PDF report generator compiles images and metadata into a PDF blob locally in the browser.
- **Audit History & Checklist State:** Stored in the browser's **localStorage** sandbox, isolated to the application origin.

Network Usage Specifications

1. **Initial Asset Download:** Internet access is required only to download the application's static HTML/JS bundles and model weights upon the first visit. Subsequent loads use the browser cache.
2. **GramLipi Translation:** To translate notices, the application makes an HTTP POST request to **http://localhost:11434/api/generate**. This is a loopback network call to an Ollama server running locally on the user's computer. No data leaves the local machine.
3. **No External Cloud Calls:** The application does not communicate with external cloud services, analytics platforms, or third-party servers during operation.